

MALON INJECTION (2ML AMP)

DESCRIPTION: MALON INJECTION is a colourless solution.

COMPOSITION: Each ml contains Metoclopramide HCl Monohydrate equivalent to Metoclopramide HCl 5 mg.

PHARMACODYNAMICS: Metoclopramide stimulates motility of the upper gastro-intestinal tract without stimulating gastric, biliary, or pancreatic secretions. Its mode of action is unclear. It seems to sensitize tissues on intact vagal innervation, but it can be abolished by anticholinergic drugs. Metoclopramide increases the tone and amplitude of gastric (especially antral) contractions, relaxes the pyloric sphincter and the duodenal bulb, and increases peristalsis of the duodenum and jejunum resulting in accelerated gastric emptying and intestinal transit. It increases the resting tone of the lower oesophageal sphincter. It has little, if any effect on the motility of the colon or gallbladder.

Metoclopramide has dopamine antagonist activity. Like the phenothiazines and related drugs which are also dopamine antagonist, Metoclopramide produces sedation and may produce extrapyramidal reactions, although these are comparatively rare (see Warnings). Metoclopramide inhibits the central and peripheral effects of apomorphine, induces release of prolactin and causes a transient increase in circulating aldosterone levels.

PHARMACOKINETICS: The onset of pharmacological action of Metoclopramide is 1 to 3 minutes following an intravenous dose, 10 to 15 minutes following intramuscular administration, and 30 to 60 minutes following an oral dose; pharmacological effects persist for 1 to 2 hours. There is marked variability in peak plasma concentrations of Metoclopramide after oral administration, which appears to be due to interindividual differences in first-pass metabolism. Plasma protein binding is 13-22%. About 80% of the drug is excreted in the urine in the first 24 hours, approximately half as the glucuronide and sulphate conjugates and half as unchanged drug. Elimination half-life varies in different studies from 2.5 to 5 hours. Impaired renal function results in reduced clearance of Metoclopramide and an increased half-life (15 hours).

INDICATION:

Adult population: Malon Injection is indicated for use in adults in:

- prevention of post-operative nausea and vomiting,
- symptomatic treatment of nausea and vomiting, including nausea and vomiting induced by migraine attacks,
- prevention of radiotherapy-induced nausea and vomiting.

Pediatric population: Malon Injection is indicated in children aged 1 to 18 years for:

- prevention of delayed chemotherapy-induced nausea and vomiting as a second line option,
- prevention of post-operative nausea and vomiting as second-line option

RECOMMENDED DOSAGE: The solution can be administered by the intravenous or intramuscular route. The intravenous doses must be administered as slow bolus (for at least 3 minutes).

All indications (adults)

A single 10 mg dose is recommended for the prevention of post-operative nausea and vomiting.

The recommended dose for the symptomatic treatment of nausea and vomiting, including nausea and vomiting induced by migraine attacks and for the prevention of radiotherapy-induced nausea and vomiting is 10 mg per dose, 1 to 3 times daily. The maximum recommended daily dose is 30 mg or 0.5 mg/kg.

Treatment duration when administering by injection should be as short as possible and a switch to administration via oral or rectal route should be instituted as quickly as possible.

All indications (children from 1 to 18 years of age)

The recommended dosage is 0.1 to 0.15 mg/kg, 1 to 3 times daily, by intravenous route.

The maximum daily dose is 0.5 mg/kg.

Dosing table:

Age	Body Weight	Dose	Frequency
1 - 3 years	10 - 14 kg	1 mg	Up to 3 times daily
3 - 5 years	15 - 19 kg	2 mg	Up to 3 times daily
5 - 9 years	20 - 29 kg	2.5 mg	Up to 3 times daily
9 - 18 years	30 - 60 kg	5 mg	Up to 3 times daily
15 - 18 years	Over 60 kg	10 mg	Up to 3 times daily

For the prevention of delayed chemotherapy-induced nausea and vomiting, the maximum treatment duration is 5 days.

For the prevention of post-operative nausea and vomiting, the maximum treatment duration is 48 hours.

Frequency of administration:

A minimum interval of 6 hours between two administrations is to be respected, even if vomiting or rejection of the dose occurs.

Special populations:

Elderly subjects: In elderly subjects, a dose reduction should be considered, based on kidney and liver function, and overall frailty.

Kidney failure: In patients with end-stage kidney failure (creatinine clearance ≤ 15 ml/min), the daily dose should be reduced by 75%.

In patients with moderate to severe kidney failure (creatinine clearance between 15 and 60 ml/min), the dose should be reduced by 50%.

Liver failure: In patients with severe liver failure, the dose should be reduced by 50%.

Other pharmaceutical forms may be more suitable for these patient populations.

Pediatric population: Metoclopramide is contraindicated in children aged less than one year.

ROUTE OF ADMINISTRATION: For Intravenous (Iv) or Intramuscular (Im) injection

CONTRAINDICATIONS:

- Hypersensitivity to the active substance or to any of the excipients listed.
- Gastrointestinal haemorrhage, mechanical obstruction or gastro-intestinal perforation for which the stimulation of gastrointestinal motility constitutes a risk.
- Confirmed or suspected pheochromocytoma, due to the risk of severe hypertension episodes.
- History of neuroleptic or metoclopramide-induced tardive dyskinesia.
- Epilepsy (increased crises frequency and intensity).
- Parkinson's disease.
- Combination with levodopa or dopaminergic agonists.
- Known history of methaemoglobinemia with metoclopramide or of NADH cytochrome-b5 deficiency
- Use in children less than 1 year of age.

WARNINGS & PRECAUTIONS:

Prolactinemia: Metoclopramide elevates prolactin levels and the elevation persists during chronic administration. Tissue culture experiments indicate that approximately one third of human breast cancers are prolactin dependent in vitro, a factor of potential importance if the prescription of Metoclopramide is contemplated in a patient with previously detected breast cancer. Although disturbances such as galactorrhoea, amenorrhoea, gynaecomastia and impotence have been reported with prolactin-elevating drugs, the clinical significance of elevated serum prolactin levels is unknown for most patients.

An increase in mammary neoplasms has been found in rodents after chronic administration of prolactin stimulating neuroleptic drugs. Neither clinical studies nor epidemiologic studies conducted to date, however, have shown an association between chronic administration of these drugs and mammary tumorigenesis; the available evidence is too limited to be conclusive at this time.

WARNING:

1. Avoid doses exceeding 0.5mg/kg/day
2. Extrapyramidal effects, especially dystonic reaction of metoclopramide are more likely to occur in children shortly after initiation of therapy, and usually with doses higher than 0.5mg per kg of body weight per day.
3. Children are particularly susceptible to the toxic effect of metoclopramide, which include extrapyramidal effects (agitation, irritability, neck pain, rigidity, dystonia).
4. Metoclopramide should not be given to children unless a clear indication has been established for its use.
5. The recommended dosage of metoclopramide should not be exceeded.

Neurological disorders: Extrapyramidal disorders may occur, particularly in children and young adults, and/ or when high doses are used. These reactions generally occur at the beginning of treatment, and can occur after a single dose. If extrapyramidal symptoms occur, metoclopramide should be discontinued immediately. These effects are generally completely reversible after treatment discontinuation; however, symptomatic treatment may be required (benzodiazepines in children, and/ or anticholinergic antiparkinsonian medicinal products in adults).

An interval of at least six hours should be respected between each dose even if vomiting or rejection of the dose occurs, in order to avoid overdose.

Long-term treatment with metoclopramide may cause potentially irreversible tardive dyskinesia, particularly in elderly subjects. Treatment should not exceed 3 months because of the risk of tardive dyskinesia. Treatment must be discontinued if clinical signs of tardive dyskinesia occur.

Neuroleptic malignant syndrome has been described with metoclopramide in combination with neuroleptics and with metoclopramide monotherapy. Metoclopramide must be immediately discontinued if symptoms of neuroleptic malignant syndrome develop and appropriate treatment should be initiated.

Particular caution should be exercised in patients with underlying neurological disorders, and in patients receiving other centrally-acting drugs.

Symptoms of Parkinson's disease may be also be exacerbated by metoclopramide.

Methemoglobinemia: Methemoglobinemia, which could be related to NADH-cytochrome b5 reductase deficiency, has been reported. If this occurs, treatment must be immediately and permanently discontinued, and appropriate measures initiated (such as treatment with methylene blue).

Cardiac disorders: Serious cardiovascular undesirable effects, including cases of severe bradycardia, circulatory collapse, cardiac arrest and Q T prolongation have been reported during administration of metoclopramide by injection, particularly via the intravenous route.

Particular caution should be exercised when administering metoclopramide, particularly via the intravenous route, in elderly subjects, patients with cardiac conduction disorders (including QT) prolongation, patients with electrolyte imbalance, bradycardia, and patients taking other drugs known to prolong QT interval.

The intravenous injection must be given as a slow bolus (of at least 3 minutes' duration) in order to reduce the risk of undesirable effects (e.g. hypotension, akathisia).

Kidney of liver failure: In patients with kidney failure of severe liver failure, a dose reduction is recommended.

Hypertension patients: It should be noted that there have been incidences of hypertension, dizziness, severe or continuing headaches in users of metoclopramide.

Mental depression: As depression is one of those rare incidences seen in takers of metoclopramide, attention should be taken when treating mental depression.

If vomiting persists in a patient receiving Metoclopramide, the patient should be reassessed to exclude the possibility of an underlying disorder e.g. cerebral irritation. Intravenous injections of Metoclopramide should be made slowly over a 1 to 2 minute period, since a transient but intense feeling of anxiety and restlessness, followed by drowsiness, may occur with rapid administration. Patients should be cautioned about engaging in activities requiring mental alertness for a few hours after the drug has been administered.

WARNING:

This preparation contains Sodium Metabisulphite that may cause allergic type reactions in certain susceptible people. Do not use if known to be hypersensitive to bisulphites.

INTERACTION WITH OTHER MEDICAMENTS:

Interactions with other drugs: The effects of Metoclopramide on gastrointestinal motility are antagonised by anticholinergic drugs and narcotic analgesics. Additive sedative effects can occur when Metoclopramide is given with alcohol, sedatives, hypnotics, narcotics or tranquillisers. Since extrapyramidal symptoms may occur with both Metoclopramide and neuroleptics such as phenothiazines, care should be exercised in the event of both drugs being prescribed concurrently.

Since Metoclopramide accelerates abnormally slow gastric and small bowel peristaltic activity, it may change absorption of orally administered drugs. The absorption of drugs from the small bowel may be accelerated (e.g. paracetamol, tetracycline, L-dopa), whereas absorption of the drugs from the stomach may be diminished (e.g. digoxin).

The finding that metoclopramide releases catecholamines in patients with essential hypertension suggest that it should be used cautiously, if at all, in patients receiving monoamine oxidase inhibitors.

Gastroparesis (gastric stasis) may be responsible for poor diabetic control in some patients. Exogenously administered insulin may begin to act before food has left the stomach and lead to hypoglycemia. Because the action of metoclopramide will influence the delivery of food to the intestines and thus the rate of absorption, insulin dosage or timing of dosage may require adjustment.

Compatibilities: Metoclopramide HCl has been stated to be compatible for up to 48 hours at room temperature, protected from light in the following solutions:

1. Dextrose 5% in Sodium Chloride 0.45%
2. Dextrose 5% in water
3. Ringer's injection
4. Ringer's injection, lactated
5. Sodium Chloride 0.9%

USE IN PREGNANCY & LACTATION:

Use in pregnancy: Animal tests in several mammalian species and clinical experience have not indicated a teratogenic effect. Nevertheless, Metoclopramide should only be used when there are compelling reasons and is not advised during the first trimester.

Use in lactation: Metoclopramide is excreted in human milk. It is not known whether it has a harmful effect on the newborn. Therefore, it is not recommended for nursing mothers unless the expected benefits to the mother outweigh any potential risk to the baby. The increased risk of adverse reactions in young children should be taken into account in making a risk benefit assessment.

SIDE EFFECTS: The most frequent adverse reactions to Metoclopramide are restlessness, drowsiness, fatigue and lassitude, which occur in approximately 10 percent of patients. Less frequently, insomnia, headache, dizziness, nausea, or bowel disturbances may occur. A single instance of supraventricular tachycardia following intramuscular administration has been reported. Although uncommon at normal dosage, various extrapyramidal reactions to Metoclopramide, usually of the dystonic type, have been reported. Reactions include: spasm of the facial muscles, trismus, rhythmic protrusion of the tongue, a bulbar type of speech, spasm of the extraocular muscles including oculogyric crises, unnatural positioning of the head and shoulders and opisthotonos. There may be a generalised increase in muscle tone. The majority of reactions occur within 36 hours of starting treatment and the effects usually disappear within 24 hours of withdrawal of the drug. However, close observation is required and in cases of more severe reactions an antiparkinson drug such as benzotropine or an anticholinergic antihistamine such as diphenhydramine should be given. Tardive dyskinesia, which may be persistent, has been reported particularly in elderly patients undergoing long term therapy with Metoclopramide.

Hepatic: Rarely, cases of hepatotoxicity characterized by such findings as jaundice and altered liver function tests, when metoclopramide was administered with other drugs with known hepatotoxic potential.

Renal: Urinary frequency and incontinence.

Hematologic: A few cases of neutropenia, leukopenia, or agranulocytosis, generally without clear-cut relationship to metoclopramide.

Allergic reactions: A few cases of rash, urticaria, or bronchospasm, especially in patients with a history of asthma. Rarely, angioneurotic edema, including glossal or laryngeal edema.

Miscellaneous: Visual disturbances. Porphyria. Rare occurrences of neuroleptic malignant syndrome (NMS) have been reported. This potentially fatal syndrome is comprised of the symptom complex of hyperthermia, altered consciousness, muscular rigidity and autonomic dysfunction.

Transient flushing of the face and upper body, without alterations in vital signs, following high doses intravenously.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: Extrapyramidal side effects are the most frequently reported adverse reactions to overdosage. Management of overdosage consists of gastric emptying, close observation and supportive therapy. Antiparkinson and antihistamine/anticholinergic drugs such as diphenhydramine hydrochloride have effectively controlled extrapyramidal reactions

INCOMPATIBILITIES : Metoclopramide HCl are stated to be incompatible with cephalothin sodium and other cephalosporins, chloramphenicol sodium and sodium bicarbonate.

Cisplatin, cyclophosphamide, doxorubicin hydrochloride and methotrexate sodium are stated to be conditionally compatible with metoclopramide hydrochloride but compatibility is dependent upon factors such as drug concentration, pH and temperature.

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. Jauhkan daripada kanak-kanak.

SHELF LIFE: Please refer to outer package.

PACK SIZE: Pack In box of 10 ampoules x 2 ml.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

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Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

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